

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 2- Relative Frequency Functions

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Session Outline

- 1 Frequency Distribution
- 2 Relative Frequency
- 3 Relative Frequency Distribution
- 4 Empirical Distribution

There are various definitions of probability. In this session, students will be introduced to relative frequency definition of probability and frequency distribution function. The empirical probability distributions and the applications of relative frequency will be introduced.

Learning Outcomes

- At the end of the session, student will be able to:
 - 1 Construct relative frequency distribution tables.
 - 2 Define probability using relative frequency.
 - 3 Construct an empirical probability distribution of a given data.
 - 4 Apply relative frequency functions in solving problems.

Frequency Distribution

- Data collected through surveys or censuses must be organized in a meaningful way so that we can use it effectively.
- One of the simple and effective way of organizing the raw data collected is the use of frequency distribution.
- **Definition:** A frequency distribution is the listing of frequencies of all categories of the observed values of a variable.
- **Alternatively,** a frequency distribution is an organized tabulation showing exactly how many individuals are located in each category on the scale of measurement.
- A frequency distribution presents an organized picture of the entire set of scores, and it shows where each individual is located relative to others in the distribution.

Frequency Distribution

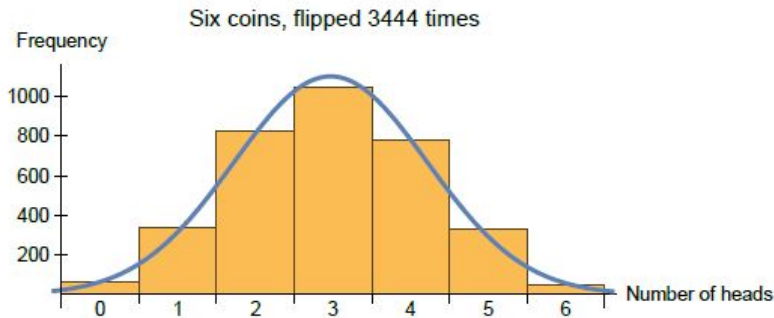
- We can construct frequency distributions for any variable. For example, consider the experiment of flipping six coins, and recording the number of heads that come up. The outcome of such an experiment will be one of the integers $\{0, 1, 2, 3, 4, 5, 6\}$.
- The experiment is repeated 3,444 times, the table below shows the frequency distribution of the result.

No. of Heads	0	1	2	3	4	5	6
Frequency	66	341	825	1048	780	333	51
Percentage	1.92	9.90	23.95	30.43	22.65	9.67	1.48

- We can compile this data into a histogram, with “frequency” on the vertical axis and “number of heads” on the horizontal axis.

Frequency Distribution

The histogram for the frequency distribution of the number of heads observed over 3,444 repetitions is presented below:



Types of Frequency Distribution

There are 3 types of frequency distributions (base on the variable):

- Categorical frequency distributions-Qualitative Variables
 - Ungrouped frequency distributions-Quantitative Variables
 - Grouped frequency distributions-Quantitative Variables
- ① Categorical frequency distributions:

Area of Study of some students in the University of Ghana

Area of Study	Frequency
Medical Sciences	15
Mathematical Sciences	23
Biological Sciences	42
Education	20
Social Sciences	52
Fine Arts	41

Types of Frequency Distribution

- ② Ungrouped frequency distributions:
Distribution of the Marks of students in Probability course.

Marks	Frequency
15	8
17	12
20	18
22	25
23	21
25	16
28	5

Types of Frequency Distribution

- 3 Grouped frequency distributions:
Distribution of the ages of employees in an IT firm.

Marks	Frequency
30-32	2
33-35	4
36-38	3
39-41	2
42-44	6
46-48	4
49-51	2

Relative Frequency

- **Definition:** A relative frequency is the ratio (fraction or proportion) of the number of times a value of the data occurs in the set of all outcomes to the total number of outcomes.
- To find the relative frequencies of a frequency distribution table, divide each frequency by the total number of observation in the sample or population.
- For example, consider twenty students who were asked how many hours they learned per day. Their responses, in hours, are as follows: 5; 6; 3; 3; 2; 4; 7; 5; 2; 3; 5; 6; 5; 4; 4; 3; 5; 2; 5; 3. Construct a relative frequency for the responses.

Relative Frequency Example

- Relative frequencies can be written as fractions, percents, or decimals.

Frequency Table of Student Work Hours with Relative Frequencies

Hours	Frequency	Relative Frequency
2	3	$\frac{3}{20} = 0.15$
3	5	$\frac{5}{20} = 0.25$
4	3	$\frac{3}{20} = 0.15$
5	6	$\frac{3}{20} = 0.30$
6	2	$\frac{2}{20} = 0.10$
7	1	$\frac{1}{20} = 0.05$

Cumulative Relative Frequency

- Cumulative relative frequency is the accumulation of the previous relative frequencies.
- Example: the table below represents the heights, in inches, of a sample of 100 male semi-professional soccer players.

Height	Freq.	Relative Freq.	Cumulative R.F
59.95-61.95	5	0.05	0.05
61.95-63.95	3	0.03	0.08
63.95-65.95	15	0.15	0.23
65.95-67.95	40	0.40	0.63
67.95-69.95	17	0.17	0.80
69.95-71.95	12	0.12	0.92
71.95-73.95	7	0.07	0.99
73.95-75.95	1	0.01	1.00
Total	100	1.00	

Relative Frequency Distribution

- **Definition:** A relative frequency distribution consists of the relative frequencies, or proportions (percentages), of observations belonging to each category.
- These relative frequencies have a useful interpretation: they provide information about the chance or probability of getting an observation from each category in a blind or random draw.
- For this reason a relative frequency distribution is often referred to as an observed or *empirical probability* distribution.

Empirical Probability Distribution

- A probability distribution obtained by means of observation and experimental methods is referred to as an empirical probability distribution , or a relative frequency distribution based on observation.
- The empirical distribution, or empirical distribution function, can be used to describe a sample of observations of a given variable.
- Its value at a given point is equal to the proportion of observations from the sample that are less than or equal to that point.
- An empirical distribution may represent either a continuous or a discrete distribution

Example

discrete description

value	probability
10	.1
20	.15
35	.4
40	.3
60	.05

continuous description

value	probability
0 – 10-	.1
10 – 20-	.15
20 – 35-	.4
35 – 40-	.3
40 – 60-	.05

Empirical Cumulative Distribution Function

- The empirical cumulative distribution function (ecdf) is closely related to cumulative frequency. Rather than show the frequency in an interval, however, the ecdf shows the proportion of scores that are less than or equal to each score
- **Definition:** Let $X_1, X_2, X_3, \dots, X_n$ be independent and identically distributed random variables, with distribution function $F(x) = P(X_i \leq x)$. The Empirical Cumulative Distribution Function (ECDF), also known simply as the empirical distribution function, is defined as

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n 1_{\{X_i \leq x\}},$$

where 1 is the indicator function, namely $1_{\{X_i \leq x\}}$ is one if $X_i \leq x$ and zero otherwise.

Example 1

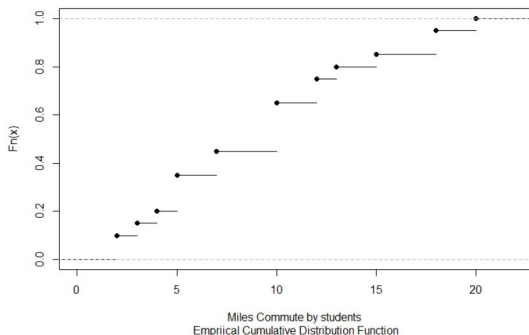
Twenty people were asked how many miles, to the nearest mile, they commute to work each day. The data are as follows: 2; 5; 7; 3; 2; 10; 18; 15; 20; 7; 10; 18; 5; 12; 13; 12; 10; 4; 5; 10.

Construct an empirical cumulative distribution for the data.

Miles Covered	Frequency	Relative Freq.	ECDF
2	2	0.10	0.10
3	1	0.05	0.15
4	1	0.05	0.20
5	3	0.15	0.35
7	2	0.10	0.45
10	4	0.20	0.65
12	2	0.10	0.75
13	1	0.05	0.80
15	1	0.05	0.85
18	2	0.10	0.95
20	1	0.05	1.00

Graph of ECDF

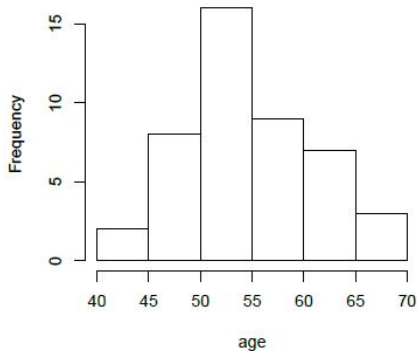
- The graph of the empirical cumulative distribution has the cumulative relative frequencies or cumulative probabilities ($F_n(x)$) on the vertical axis and the data values (class boundaries or midpoints) on the horizontal axis.
- From example 1 above, the empirical cumulative distribution function graph is as below:



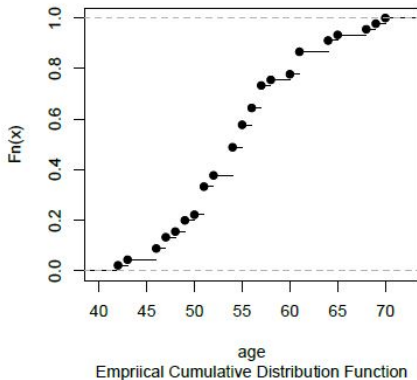
An empirical cumulative distribution function and histogram of the ages of the presidents of the United States at the time of their inauguration is given below:

(57,61,57,57,58,57,61,54,68,51,49,64,50,48,65,52,56,46,54,49,51,47,55,55,54,42,51,56,55,51,54,51,60,61,43,55,56,61,52,69,64,46,54,47,70)

Histogram of age



Age of Presidents at Inauguration



Trial Questions

Suppose that a new AIDS antibody drug is currently under study. It is given to patients once the AIDS symptoms have revealed themselves. Of interest is the average (mean) length of time in months patients live once starting the treatment. Two researchers (A and B) each follow a different set of 40 AIDS patients from the start of treatment until their deaths. The following data (in months) collected are presented below. Using this information to:

- 1 construct relative frequency and empirical cumulative distribution function for each researcher.
- 2 sketch the histogram and empirical cumulative distribution function for the two dataset.

Note: Use the class boundaries: 0.5-6.5, 6.5-12.5, 12.5-18.5, 18.5-24.5, ...

Trial Questions Cont'd.

A	A	A	A		B	B	B	B
3	14	13	31		3	14	29	23
4	24	21	34		14	14	13	21
11	25	22	29		11	26	18	21
15	15	10	17		5	25	24	16
16	26	12	8		16	21	42	12
17	27	8	24		17	22	33	18
22	33	40	18		28	31	29	41
44	29	32	47		41	25	2	22
37	35	26	33		31	33	35	16
16	44	27	34		18	34	44	23

THANK YOU!!!